

Dilip Kumar R

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Education

B-Tech COMPUTER SCIENCE AND ENGINEERING

CMR University, Bangalore.

September 2023 - August 2027 *expected

CGPA (current) : 8.86

Pre University

St. Joseph's Pre University College, Bangalore.

July 2021 - March 2023

Grade : 77%

Experience

SCIENTIST/ENGINEER INTERN

Indian Space Research Organisation

October 2025 - November 2025

Bangalore, India

- Revamped flight software telemetry verification system using **Python**, slashing verification time by **50%** through the elimination of manual, hard-coded checks, and validating **50+** telemetry parameters.
- Developed a production-grade **Python** library to parse **Ada**-based flight software and extract telemetry parameters, enabling automated data validation and visualization workflows.
- Wrote a comprehensive test suite with **>65%** code coverage using **pytest** and **unittest**, implementing unit tests that reduced debugging time by **40%**.

Open Source Contributions

AsyncApi:

- Enhanced the template quality by replacing direct calls to the server with helper functions in the template, resulting in a performance enhancement of over 20%. pr: [#1632](#)

Kubeflow:

- Redesigned the navigation bar with modern UI, full mobile responsiveness using Tailwind CSS utility classes and media queries, significantly improving overall user experience. pr: [#4270](#)
- Integrated the Pipelines and Components into the Kubeflow SDK, allowing users to install dependencies in one command.

Projects

Freddy Bot *AI analysis, Github Repository Analysis, Telegram integration.*

[\(live-link\)](#)

[Github-Link](#)

- Constructed an open-source developer productivity tool that delivers **real-time Telegram** notifications to repository contributors when new issues are created, streamlining response times.
- Architected an intelligent automation pipeline leveraging **HuggingFace** and open source models like **Mistral-7B** and **Llama-3B** to generate contextual and rapid summaries, including the solution for the issue in the repository. Resulting in a significant increase in contributor productivity.

Tech Stack: Typescript, Python, HuggingFace, Github API, Telegraf, React.

Glyph *AWS, Agentic Orchestration, Code generation, Langchain.*

[Github-Link](#)

- Engineered an **Agentic Orchestration** layer for Manim code generation from **GLM-4** and **Qwen-coder2.5**, automated S3 uploads, and optimised output videos using **Ffmpeg** compression algorithms.
- Built a **feedback loop** that significantly reduced LLM hallucinations in code generation while preserving the user's original prompt context.
- Implemented a **Human-in-the-Loop** feedback system with dynamic **flowchart generation** to visualize logic before execution, improving the accuracy and reliability of LLM-generated code.
- Integrated a **Retrieval-Augmented Generation (RAG)** pipeline to enhance contextual grounding for general conversational interactions.

Tech Stack: Python, Langchain, Langgraph, Huggingface, Boto3, Manim, Ffmpeg.

Technical Skills

Languages : C, C++, Ada, Python, JavaScript, Typescript.

Frameworks : React js, Node js, Express js, MongoDB, LangChain, LangGraph, PyTorch.

Technologies : Git, Github, Docker, AWS.

Achievements & Certifications

- Winner - DevStorm Hackathon
- 3rd Runner up - MarchMania hackathon
- 1400+ rating in Leetcode
- 1000+ rating in Codeforces
- Coursera : Machine Learning Specialisation - Andrew Ng
- DeepLearningAI : Langgraph: Building AI agents with OpenAI